

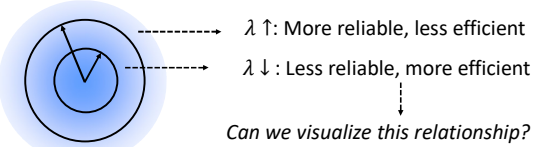
I. Introduction

Contextual robust optimization is a standard framework for decision-making under uncertainty.

$$\min_{z \in \mathcal{Z}} \max_{y \in U_\lambda(x)} f(z, y)$$

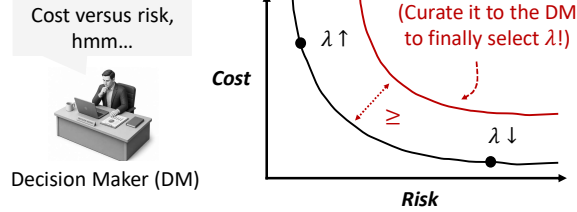
Decision variable Adversarial variable Robustness parameter e.g., radius

A Fundamental Question: How to select λ ?



Our Objective: Robustness Calibration

Using data, estimate a **certified** Pareto frontier that conservatively represents the tradeoff between the cost and risk of the RO for different selections of λ .



II. Preliminary: Conformal Risk Control [1]

Extends standard CP from controlling miscoverage to any risk function.

Objective: Given risk bound α , select loss function parameter λ such that

$$\mathbb{E}[\ell_\lambda(X_{n+1}, Y_{n+1})] \geq 1 - \alpha \quad (1)$$

Estimator: Let $\bar{\ell}_n(\lambda) = \frac{1}{n} \sum_{i=1}^n \ell_\lambda(x_i, y_i)$ denote the sample average loss

$$\hat{\lambda} = \inf_{\lambda \in \Lambda} \left\{ \lambda : B - \bar{\ell}_n(\lambda) \geq \frac{[(n+1)(B-\alpha)]}{n} \right\}$$

Where B denotes the essential supremum of the loss function.

Example: $B = 1$ and $\ell_\lambda(x, y) = \mathbf{1}\{Y \in C(x)\}$ recovers standard CP.

Theorem (Validity): $\hat{\lambda}$ satisfies the objective eq. (1).

III. Methodology

Step 1 (Pareto Frontier Estimation):

Definition: Inverse Conformal Risk Control Estimator

Given desired robustness parameter λ , output risk upper bound:

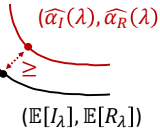
$$\hat{\alpha} = \inf_{\alpha \in [0,1]} \left\{ \alpha : B - \bar{\ell}_n(\lambda) \geq \frac{[(n+1)(B-\alpha)]}{n} \right\} \quad (2)$$

Both cost and risk can be formalized by some notion of loss functions:

$$\text{Cost} \leftarrow R_\lambda = f(z^{\text{RO}}(\lambda); Y) - f(z^*; Y)$$

$$\text{Risk} \leftarrow I_\lambda = \mathbf{1}\{Y \notin U_\lambda\}$$

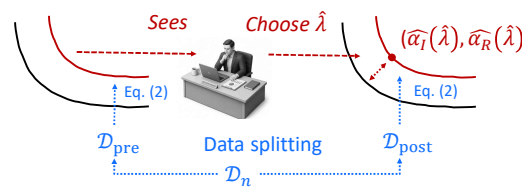
For each λ , use eq. (2) to construct estimators $\hat{\alpha}_I(\lambda)$ and $\hat{\alpha}_R(\lambda)$ to form the Pareto frontier



Step 2 (Post-hoc Selection):

Choose $\hat{\lambda}$ from $(\hat{\alpha}_I(\lambda), \hat{\alpha}_R(\lambda))$

To maintain validity, this step uses another splitted calibration dataset

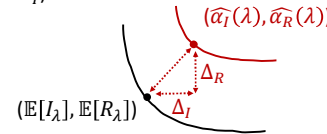


IV. Theoretical Results

Denote the gap by Δ (either Δ_R or Δ_I):

Theorem: Validity

Under exchangeability,
 $\mathbb{E}[\Delta] \geq 0$



Theorem: Error Bound

Under i.i.d.:

$$|\Delta| \leq O\left(\frac{B\sqrt{n}}{n+1}\right)$$

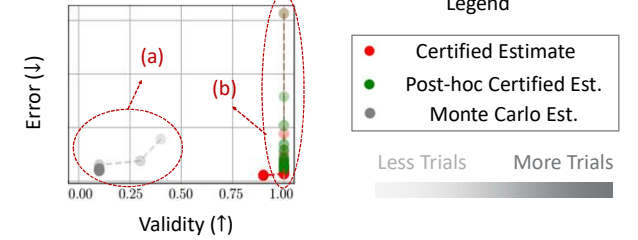
Theorem (Informal): Pareto Frontier

Under the majorant consistency assumption, the curve of $(\mathbb{E}[I_\lambda], \mathbb{E}[R_\lambda])$ by varying λ is weakly decreasing.

VI. Experiment Results

Exp 1: Estimation Quality Evaluation

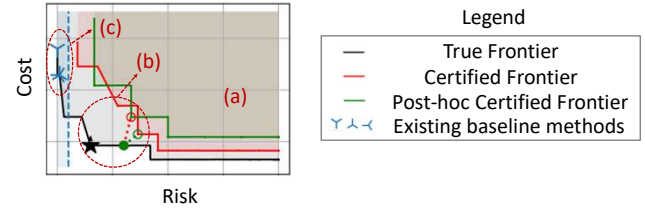
$\hat{\alpha}(\lambda)$ Evaluation Results (LP)



Observations: (a) Baseline achieves only $\leq 50\%$ Valid, (b) Our methods achieve $\approx 100\%$ Valid + Asymptotic consistency.

Exp 2: Estimation Quality Evaluation

Constructed Frontier (LP)



Observations: (a) Our constructed Pareto frontier is certified, (b) Baselines only focus on satisfying low risk, (c) In contrast, our frontier delivers useful information for the cost-risk tradeoff, and supports accurate optimal tradeoff point selection.

Footnotes:

- See the paper for more optimization examples beyond LPs.
- Our experiment uses a more computationally efficient estimator:

$$\bar{\alpha}_\epsilon(\lambda) = \frac{n}{n+1} \bar{\ell}_n(\lambda) + \frac{B}{n+1}$$

which is a lower bound proxy to the proposed estimator in eq. (2).

VII. Reference

- [1] Angelopoulos, A., Bates, S., Fisch, A., Lei, L., & Schuster, T. (2024, May). Conformal risk control. In International conference on learning representations (Vol. 2024, pp. 55198-55218).